

Numerical Analysis – Lecture 6

The Givens algorithm Given $m \times n$ matrix A , let ℓ_i be the number of leading zeros in the i th row of A , $i = 1, 2, \dots, m$.

Step 1 Stop if the (integer) sequence $\{\ell_1, \ell_2, \dots, \ell_m\}$ increases monotonically, the increase being strictly monotone for $\ell_i \leq n$.

Step 2 Pick any two integers $1 \leq p < q \leq m$ such that either $\ell_p > \ell_q$ or $\ell_p = \ell_q < n$.

Step 3 Replace A by $\Omega^{[p,q]}$, a Givens rotation that annihilates the $(q, \ell_q + 1)$ element.

Update the values of ℓ_p and ℓ_q and go to *Step 1*.

The final matrix A is upper triangular and also has the property that the number of leading zeros in each row increases *strictly monotonically* until all the rows of A are zero – a matrix of this form is said to be in *standard form*. This end result, as we recall, is the required matrix R .

The cost There are less than mn rotations and each rotation replaces two rows by their linear combinations, hence the total cost is $\mathcal{O}(mn^2)$.

If we wish to obtain explicitly an orthogonal Q s.t. $A = QR$ then we commence by letting Ω be the $m \times m$ unit matrix and, each time A is premultiplied by $\Omega^{[p,q]}$, we also premultiply Ω by the same rotation. Hence the final Ω is the product of all the rotations, in correct order, and we let $Q = \Omega^T$. The extra cost is $\mathcal{O}(m^2n)$. However, in most applications we don't need Q but, instead, just the action of Q^T on a given vector (recall: solution of linear systems). This can be accomplished by multiplying the vector by successive rotations, the cost being $\mathcal{O}(mn)$.

4.6 Householder rotations

Let $\mathbf{u} \in \mathbb{R}^m \setminus \{\mathbf{0}\}$. The $m \times m$ matrix $I - 2\frac{\mathbf{u}\mathbf{u}^T}{\|\mathbf{u}\|^2}$ is called a *Householder rotation*. Each such matrix is symmetric (that's trivial) and orthogonal, since

$$\left(I - 2\frac{\mathbf{u}\mathbf{u}^T}{\|\mathbf{u}\|^2}\right)^T \left(I - 2\frac{\mathbf{u}\mathbf{u}^T}{\|\mathbf{u}\|^2}\right) = \left(I - 2\frac{\mathbf{u}\mathbf{u}^T}{\|\mathbf{u}\|^2}\right)^2 = I - 4\frac{\mathbf{u}\mathbf{u}^T}{\|\mathbf{u}\|^2} + 4\frac{\mathbf{u}(\mathbf{u}^T\mathbf{u})\mathbf{u}^T}{\|\mathbf{u}\|^4} = I.$$

Deriving the first column of R Our goal is to multiply an $m \times n$ matrix A by a sequence of Householder rotations so that each product 'peels off' requisite nonzero elements in a whole successive column. To start with, we seek a rotation that transforms the first nonzero column of A to a multiple of $\mathbf{e}_1$.

Let $\mathbf{a} \in \mathbb{R}^m$ be the first nonzero column of A . We wish to choose $\mathbf{u} \in \mathbb{R}^n$ s.t. the last $m-1$ entries of $\left(I - 2\frac{\mathbf{u}\mathbf{u}^T}{\|\mathbf{u}\|^2}\right)\mathbf{a} = \mathbf{a} - 2\frac{\mathbf{u}^T\mathbf{a}}{\|\mathbf{u}\|^2}\mathbf{u}$ vanish and, in addition, we normalize $\mathbf{u}$ so that $2\mathbf{u}^T\mathbf{a} = \|\mathbf{u}\|^2$ (recall that $\mathbf{a} \neq \mathbf{0}$). Therefore $u_i = a_i$, $i = 2, \dots, m$ and the normalization implies that

$$2u_1a_1 + 2\sum_{i=2}^m a_i^2 = u_1^2 + \sum_{i=2}^m a_i^2 \Rightarrow u_1^2 - 2u_1a_1 + a_1^2 - \sum_{i=1}^m a_i^2 = 0 \Rightarrow u_1 = a_1 \pm \|\mathbf{a}\|.$$

It is usual to let the sign be the same as the sign of a_1 .

For large m we do not execute explicit matrix multiplication. Instead, to calculate $\left(I - 2\frac{\mathbf{u}\mathbf{u}^T}{\|\mathbf{u}\|^2}\right)A = A - 2\frac{\mathbf{u}(\mathbf{u}^T A)}{\|\mathbf{u}\|^2}$, first evaluate $\mathbf{w}^T := \mathbf{u}^T A$, subsequently forming $A - \frac{2}{\|\mathbf{u}\|^2}\mathbf{u}\mathbf{w}^T$.

Next columns of R Suppose that $\mathbf{a}$ is the first column of A that isn't compatible with standard form (previous columns have been, presumably, already dealt with by Householder rotations) and that the standard form requires to bring the $k+1, \dots, m$ components to zero. Hence, nonzero elements in previous columns must be confined to the first $k-1$ rows and we want them to be unamended by the rotation. Thus, we let the first $k-1$ components of $\mathbf{u}$ be zero and choose $u_k = a_k \pm (\sum_{i=k}^m a_i^2)^{1/2}$ and $u_i = a_i, i = k+1, \dots, m$.

The Householder method We process columns of A in sequence, in each stage premultiplying a current A by the requisite Householder transformation. The end result is an upper triangular matrix R in its standard form.

Example

$$A = \begin{bmatrix} 2 & 4 & 7 \\ 0 & 3 & -1 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & -2 \end{bmatrix} \Rightarrow \mathbf{u} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ 1 \\ -2 \end{bmatrix} \Rightarrow \left(I - 2 \frac{\mathbf{u}\mathbf{u}^T}{\|\mathbf{u}\|^2} \right) A = \begin{bmatrix} 2 & 4 & 7 \\ 0 & 3 & -1 \\ 0 & 0 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Calculation of Q If the matrix Q is required in an explicit form, set $\Omega = I$ initially and, for each successive rotation, replace Ω by

$$\left(I - 2 \frac{\mathbf{u}\mathbf{u}^T}{\|\mathbf{u}\|^2} \right) \Omega = \Omega - \frac{2}{\|\mathbf{u}\|^2} \mathbf{u}(\mathbf{u}^T \Omega).$$

As in the case of Givens rotations, by the end of the computation, $Q = \Omega^T$. However, if we require just the vector $\mathbf{c} = Q^T \mathbf{b}$, say, rather than the matrix Q , then we set initially $\mathbf{c} = \mathbf{b}$ and in each stage replace $\mathbf{c}$ by

$$\left(I - 2 \frac{\mathbf{u}\mathbf{u}^T}{\|\mathbf{u}\|^2} \right) \mathbf{c} = \mathbf{c} - 2 \frac{\mathbf{u}^T \mathbf{c}}{\|\mathbf{u}\|^2} \mathbf{u}.$$

Deciding between Givens and Householder rotations If A is dense, it is in general more convenient to use Householder rotations. Givens rotations come into their own, however, when A has many leading zeros in its rows. In an extreme case, if an $n \times n$ matrix A consists of zeros underneath the first subdiagonal, they can be 'rotated away' in $n-1$ Givens rotations, at the cost of just $\mathcal{O}(n^2)$ operations!